Array Matrix

Objective

So far, we have learned how to write programs that involve conditionals and iterations. Towards the very beginning of the semester, we looked at variables and how they store values we need for calculations. This week, we will look at a special kind of variable – one that lets us store and access several values under the same name. These types of variables are called **arrays**. In today's lab, we will learn more about arrays and then write a few programs using arrays.

Exercise Set A

Task 1:

Sum of Array Elements

```java
import java.util.Scanner;

public class ArraySum {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Step 1: Take size of array
        System.out.print("Enter number of elements: ");
        int n = sc.nextInt();

        // Step 2: Declare array
        int[] arr = new int[n]; //int arr[] = new int[n];

        // Declare and initialize array
        // int[] arr = {1, 2, 3, 4, 5};

        // Step 3: Take input
        System.out.println("Enter " + n + " elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        // Step 4: Calculate sum
        int sum = 0;
        for (int i = 0; i < n; i++) {
            sum = sum + arr[i];
        }

        // Step 5: Output
        System.out.println("Sum = " + sum);
    }
```

```
> run ArraySum
Enter number of elements: 3
Enter 3 elements:
5
7
-2
Sum = 10
```

Task 2:

Using Arrays Class

```java
import java.util.Arrays;

public class ArraysExample {
    public static void main(String[] args) {

        // Declare and initialize array
        int[] arr = {5, 2, 9, 1, 6};

        // Print array (without loop) using Arrays.toString()
        System.out.println("Original array: " + Arrays.toString(arr));

        // Print array with loop
        System.out.print("Printing array using loop: ");
        for (int i = 0; i < arr.length; i++) {
            System.out.print(arr[i]+" ");
//            if (i < arr.length - 1) {
//                System.out.print(", ");
//            }
        }

        // Sort the array
        Arrays.sort(arr);
        System.out.println("\nSorted array: " + Arrays.toString(arr));
    }
}
```

```
> run ArraysExample
Original array: [5, 2, 9, 1, 6]
Printing array using loop: 5 2 9 1 6
Sorted array: [1, 2, 5, 6, 9]
```

Exercise Set B

Array

Task 3:

Write a java program that reads 10 numbers from the user, and then prints them in the reverse order.

Task 4:

Write a java program that reads 20 numbers from the user, and then prints **only the even numbers** in reverse order.

Task 5:

Write a java program that reads 10 numbers from the user. Then read another number from the user, and print "yes" if this number exists among the first 10. Print "no" otherwise.

Task 6:

Write a java program that reads 15 numbers from the user, all the numbers within the range 0-9. Then print the number of times each number has been entered by the user.

Task 7:

Write a java program that reads 10 numbers from the user. After reading each number, print all the numbers that have been entered so far.

Task 8:

Write a java program that reads 10 numbers from the user, but does not allow the user to enter duplicates. This means that if a number has been entered already, the program will not accept it as input again and instead ask the user to enter a different number.

Task 9:

Write a java program that reads 10 numbers from the user. Write the program in such a way so that

if the user enters 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, then the output should be 1, 3, 5, 7, 9, 2, 4, 6, 8, 10.

If the user enters 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, then the output should be 10, 30, 50, 70, 90, 20, 40, 60, 80, 100.

If the user enters 2, 5, 6, 9, 12, 13, 14, 15, 16, 17 then the output should be 2, 6, 12, 14, 16, 5, 9, 13, 15, 17.

Task 10:

Write a Java code of a function that takes an array of integers (of positive and negative numbers) and the length or size of the array as parameters, then modifies the array by getting rid of the negative elements (the numbers after the removed one will have to shift forward to fill the gap of course). If there are multiple negative elements, remove all of those. The function returns the new logical size of the array. (New logical size represents the number of non-negative elements remaining in the array).

Sample Input/Output:

```
Original array:
3 -2 5 -1 8  Original size = 5
Updated array:
3 5 8  New size = 3
```

Exercise Set C

Matrix

Task 11:

Write a java program that creates an M by N matrix, where M and N are values given by the user. The user will provide values for the matrix. Print the values in the matrix (each of the M rows should be on separate lines) to the standard output.

Obviously you need to use a 2 dimensional array here to keep the matrix values.

Task 12:

Write a java program that creates an M by N matrix of 0's, where M and N are provided by the user.

In other words, you create a 2D array with the given dimensions, and then print it out. It is a matrix of all zeros, but if you remember that integer and double array is initialized with 0, nothing more needs to be done here.

Task 13:

Write a java program that creates an M by N matrix from a 2D array provided by the user.

Assume the following exists in your main method:

double array[][]={{1.0,2.0,3.0},{3.0,1.0,2.5},{1.5,6.5,7.25},{5.5,6.5,7.5}};

There is a 2D array provided, or you take one as input from the user, and then copy the data from that to a matrix (which is off course just another 2D array :P)

Task 14:

Write a java program that creates an M by N matrix, where M and N will be given by the user, containing random values between 0 and 1.

You need to make use of a random number generator here that generates values between 0 and 1 and assign that to the cells of the matrix. Find out how to generate random numbers from the internet. Note that your matrix should be able to hold real numbers as well as integers.

Task 15:

Write a java program that creates the N by N identity matrix where N is given by the user.

You should already know from high school math what an identity matrix is. If you do not know, or you have forgotten, then Google is your friend.

Task 16:

Write a java program that, given an M x N matrix, swap the rows i and j, where i and j are provided by the user

Assuming that there is matrix (i.e. you create one with hard coded values or better still take one from the user as input) you ask the user to enter the row numbers to be swapped … ;-) you should be able to figure out the rest :P

Task 17:

Write a java program that, given an M x N matrix, creates the transpose of the matrix.

If you do not know what the transpose of a matrix is, take a look at http://www.mathwords.com/t/transpose_of_a_matrix.htm

Task 18:

Write a java program that, given two M x N matrices, adds them.

Note that for this and all questions before this, it is up to you to check the validity of your input, i.e. before actually adding the two matrices, you need to check their dimensions to see if the can actually be added.

Task 19:

Write a java program that, given two M x N matrices, subtracts the second one from the first.

Task 20:

Write a java program that, given two matrices, check if they are equal.

Here you need to compare two matrices value by value and find out if they are equal or not.

Task 21:

Write a java program that, given two matrices, finds out their product (first one times second one)

If you forgot how matrices are multiplied, again Google is your friend.